

Test Project Outline – Module B – REST API Client Frontend REST API 客戶端 前端

Competition time

Competitors will have **3 hours** to complete Module B.

選手有 3 小時完成模組 B

Introduction 介紹

Shanghai Pudong International Airport (PVG) operates **ReClaim**, a single lost-property service for the whole airport. Passengers use ReClaim to report items they have lost, follow the progress of their claims, withdraw claims when an item is found elsewhere, and maintain their contact details.

上海浦東國際機場（PVG）營運 **ReClaim**，一套涵蓋整座機場的單一失物招領服務。旅客使用 ReClaim 來申報遺失的物品、追蹤申請進度、在物品於他處尋獲時撤回申請，以及維護自己的聯絡資料

In this module, you must build the **ReClaim passenger portal**, a consumer-facing frontend for the provided ReClaim REST API. You are responsible for the client-side user interface and user experience only. The staff desk and administration interface are outside the scope of this module.

在本模組中，你必須建構 **ReClaim 旅客入口網站**，這是為所提供的 ReClaim REST API 打造的、面向消費者的前端。你只負責客戶端的使用者介面與使用者體驗。專員櫃檯與後台管理介面不在本模組範圍內

General Description of Project and Tasks 專案與任務的一般說明

You will be given a working solution of the Module A ReClaim API. You must use the provided solution and must not implement, replace, or modify the backend. The API documentation supplied with the solution is the source of truth for request fields, response structures, validation rules, status codes, and business rules.

你將取得一份可運作的模組 A ReClaim API 解決方案。你必須使用所提供的解決方案，且不得實作、替換或修改後端。隨解決方案提供的 API 文件是請求欄位、回應結構、驗證規則、狀態碼與商業規則的權威依據（source of truth）

Assessment aid in the provided API: GET /passenger/claims?status=rejected is

intentionally delayed by about **2 seconds**. Other status filters respond normally. This delay exists only so out-of-order responses can be tested reliably. Do not special-case only rejected in your client; handle stale responses for every filter change.

所提供 API 中的評分輔助設計：`**`GET /passenger/claims?status=rejected`` 被刻意延遲約 `**2 秒**`。其他狀態篩選正常回應。此延遲的存在，只是為了能可靠地測試亂序回應的處理。請勿在你的客戶端中只針對 ``rejected`` 做特例處理；請對每一次篩選變更都處理過時（stale）的回應

Create the application as a **Single Page Application (SPA)** using a modern JavaScript framework. Additional libraries may be used. Routing must be managed by the framework, and reloading a route must restore the same page, except for unsaved form input and temporary messages.

請以現代 JavaScript 框架將應用程式建構為`**單頁應用程式（SPA）**`。可額外使用其他函式庫。路由必須由框架管理，且重新載入某路由必須還原到相同的頁面，唯未儲存的表單輸入與暫時性訊息除外

Your API base URL: `https://module-a-solution-wsYY-YYYY.foredu.cn/api`

All API paths in this document are relative to this base URL.

本文件中所有 API 路徑皆相對於此 base URL

The OpenAPI documentation of the backend API is available in `assets/api-docs/` — open `assets/api-docs/index.html` in a browser.

後端 API 的 OpenAPI 文件位於 ``assets/api-docs/`` — 在瀏覽器中開啟 ``assets/api-docs/index.html``

A runnable copy of the backend is also provided: unzip `assets/solution-api.zip`, import `assets/database/reclaim-db.sql` into MySQL, set the database credentials in `api/.env`, and serve it (see `RUN.md` inside the zip). It behaves identically to the hosted API, including the intentional rejected filter delay.

亦提供一份可執行的後端副本：解壓縮 ``assets/solution-api.zip``，將 ``assets/database/reclaim-db.sql`` 匯入 MySQL，於 ``api/.env`` 設定資料庫憑證，並啟動服務（見 zip 內的 ``RUN.md``）。其行為與託管版 API 完全相同，包含刻意設計的 ``rejected`` 篩選延遲

An extra helper endpoint is available: **POST /reset-db**. You may call it to reset the

database to the canonical seed data when needed during development or testing. It is not part of the passenger portal UI you must build.

另提供一個額外的輔助端點：`**POST /reset-db**`。你可在開發或測試期間視需要呼叫它，將資料庫重設為標準的種子資料。它不屬於你必須建構的旅客入口網站 UI

Only the following API operations are in scope:

僅以下 API 操作在範圍內:

Area	API operations
Passenger authentication	POST /passenger/register, POST /passenger/login, POST /passenger/logout
Passenger claims	GET /passenger/claims, GET /passenger/claims/{id}, POST /passenger/claims, POST /passenger/claims/{id}/withdraw
Passenger profile	GET /passenger/profile, PUT /passenger/profile
Public tracking	GET /claims/track/{referenceCode}
Terminal selection	GET /terminals

Do not call or create interfaces for staff authentication, terminal management, found items, staff claim processing, matching or resolution, dashboards, or GET /my-terminals. GET /terminals may only be used as a read-only source for the terminal selector in the new-claim form.

請勿呼叫或建立以下介面：專員驗證、航廈管理、拾獲物品、專員端的申請處理、配對或結案、儀表板，或 ``GET /my-terminals``。``GET /terminals`` 只能作為新申請表單中航廈選擇器的唯讀資料來源使用

The following passenger accounts are available for testing:

以下旅客帳號可供測試:

Email	Password	Notes
passenger1@email.com	passenger123	Active passenger
passenger2@email.com	passenger123	Active passenger
passenger31@email.com	passenger123	Deactivated account; login must fail

Requirements 需求

Authentication 驗證

The portal must use the passenger authentication system. An authenticated request must include the passenger token as:

入口網站必須使用旅客驗證系統。已驗證的請求必須將旅客 token 以下列方式帶入:

Authorization: Bearer {token}

The session must survive page reloads. On sign-out, or when a request that requires authentication receives a 401 Unauthenticated response, the current session must be cleared. Only passenger credentials and tokens may be used; credentials and tokens of staff or system operators with elevated privileges must not.

工作階段必須能在頁面重新載入後存活。在登出時，或當需要驗證的請求收到`401 Unauthenticated` 回應時，必須清除目前的工作階段。只能使用旅客的憑證與 token；不得使用具有較高權限的專員或系統操作員的憑證與 token

Application shell and navigation 應用程式外殼與導覽

The application must have a clear, consistent passenger-facing layout and navigation. Visual design polish is not the most important factor; a working, usable application that correctly supports the required passenger workflows is. A Font Awesome package is provided in assets/fontawesome for icons if you choose to use it.

應用程式必須具備清楚、一致的、面向旅客的版面與導覽。視覺設計的精緻度

並非最重要的因素；一個能正確支援所需旅客工作流程、可運作且易用的應用程式才是重點。`assets/fontawesome` 提供了 Font Awesome 套件，若你選擇使用可用於圖示

Use path-based client routes (framework History mode). These paths are mandatory for assessment — markers may open them directly. Do not use hash routing (`#/...`).

使用**以路徑為基礎**的客戶端路由（框架的 History 模式）。這些路徑為評分所必需—評分者可能會直接開啟它們。請勿使用 hash 路由（`#/...`）

- Unauthenticated visitors must be able to register, sign in, and use public claim tracking.

未驗證的訪客必須能夠註冊、登入，並使用公開的認領追蹤

- Authenticated passengers must additionally be able to file a claim, view their claims, open a claim's details, edit their profile, and sign out.

已驗證的旅客必須另外能夠提交申請、檢視自己的申請、開啟某筆申請的詳情、編輯個人資料，以及登出

- Authenticated routes must be guarded. If an unauthenticated visitor requests one, navigate them to the sign-in page and return them to the originally requested URL after successful authentication.

需驗證的路由必須被阻擋。若未驗證的訪客請求此類路由，將其導向登入頁，並在成功驗證後把他們帶回原本請求的 URL

- The default route `/` must redirect authenticated passengers to the My claims page and others to the sign-in page.

預設路由 `/` 必須將已驗證的旅客重新導向至「我的申請」頁面，其他人則導向登入頁

- Controls and navigation for staff or administrative functions must not appear.

不得出現任何專員或後台管理功能的控制項與導覽

- The interface must make these situations visually and textually distinct so the passenger always understands what is happening:

介面必須在視覺與文字上明確區分以下情境，讓旅客隨時都清楚正在發

生什麼事:

- **Loading** — data or an action is in progress (for example a spinner, skeleton, or “Loading...” message). Do not show an empty list or a success message while waiting.

載入中— 資料或某項動作正在進行中（例如旋轉圖示、骨架畫面或「Loading…」訊息）。等待期間請勿顯示空清單或成功訊息

- **Empty** — the request succeeded but there is nothing to show (for example “You have no claims yet”). This must not look like an error or like loading.

空— 請求成功但沒有可顯示的內容（例如「你尚無任何申請」）。它不得看起來像錯誤或載入中

- **Success** — an action completed correctly (for example a confirmation after filing a claim or updating the profile).

成功— 某項動作正確完成（例如提交申請或更新個人資料後的確認）

- **Error** — a request failed or validation prevented submission; show a clear message and, where useful, a way to retry.

錯誤— 請求失敗或驗證阻擋了提交；顯示清楚的訊息，並在有幫助時提供重試的方式

Registration page 註冊頁面

The registration page route is /register. It must create a passenger account.

註冊頁路由為 `register`。它必須能建立旅客帳號

The form must collect:

表單必須蒐集:

- First name
- Last name
- Email address
- Phone number (optional)
- Address line 1

- Address line 2 (optional)
- City
- Postcode
- Country
- Password, with a minimum length of 8 characters

Validate required fields, email format, and password length before submission. Display field-specific validation messages returned by the API. A successful registration returns a passenger token and passenger details; store the session and navigate to the My claims page.

在提交前驗證必填欄位、email 格式與密碼長度。顯示 API 回傳的、針對特定欄位的驗證訊息。註冊成功會回傳旅客 token 與旅客詳情；儲存工作階段並導向「我的申請」頁面

Sign-in page and sign out 登入頁與登出

The sign-in page route is /login. It must accept an email address and password and authenticate through the appropriate endpoint.

登入頁路由為 `/login`。它必須接受 email 地址與密碼，並透過適當的端點進行驗證

- On success, store the returned passenger token and passenger details and continue to the originally requested authenticated URL, or to the My claims page if none was requested.

成功時，儲存回傳的旅客 token 與旅客詳情，並繼續前往原本請求的需驗證 URL；若無，則前往「我的申請」頁面

- Invalid credentials or an inactive account return 401; show a clear message without exposing sensitive information.

憑證無效或帳號未啟用時回傳 `401`；顯示清楚的訊息，且不得洩漏敏感資訊

- An authenticated passenger must be able to sign out from the application shell.

已驗證的旅客必須能從應用程式外殼登出

- Signing out must clear the local session and navigate to the sign-in page. The local session must still be cleared if the remote token is already invalid.

登出必須清除本機工作階段並導向登入頁。即使遠端 token 已失效，本機工作階段仍必須被清除

New claim page 新申請頁

The new-claim page route is /claims/new. Authenticated passengers must be able to report a lost item here.

新申請頁路由為 `/claims/new`。已驗證的旅客必須能在此申報遺失物品

Implement claim reporting as an **assisted multi-step process**:

以**輔助式多步驟流程**實作申請的申報:

1. **Where and when** — select the terminal and date lost.
地點與時間** — 選擇航廈與遺失日期
2. **Item details** — enter the category, brand, colour, description, and flight number.
物品細節** — 輸入類別、品牌、顏色、描述與航班編號
3. **Review and submit** — show a complete summary and allow the passenger to return to either previous step without losing data.
檢視與提交** — 顯示完整摘要，並讓旅客能在不遺失資料的情況下返回前面任一步驟

The journey must show the current step and completed steps. Passengers may move forward only when the current step is valid. Browser back and forward navigation must not accidentally submit the form or lose already entered values. 流程必須顯示目前步驟與已完成步驟。旅客只有在目前步驟有效時才可往前。瀏覽器的上一頁／下一頁導覽不得意外提交表單或遺失已輸入的值

The form must contain:

表單必須包含:

Field	Requirement
Terminal	Required
Category	Required. Use exactly: electronics, documents, luggage, clothing, jewellery, keys, other.
Brand	Optional
Colour	Optional
Description	Required
Date lost	Required. Submit as lost_on; a future date must not be accepted.
Flight number	Optional

Client-side validation must prevent clearly invalid submissions. API validation errors must be shown beside the relevant fields where possible.

客戶端驗證必須防止明顯無效的提交。API 的驗證錯誤必須盡可能顯示於相關欄位旁

An unfinished claim draft must be recoverable after a page reload or when the browser is reopened: offer to continue it or discard it. A previous draft must not be retrieved if it was already submitted or discarded, or if a different passenger is signed in.

未完成的申請草稿必須能在頁面重新載入後或瀏覽器重新開啟後復原：提供「繼續」或「捨棄」的選項。若草稿已提交或已捨棄，或若目前登入的是另一位旅客，則不得取回先前的草稿

While terminals are loading, show a loading state and do not allow the passenger to continue without a selected terminal. If loading fails, show an error and a retry

action. The passenger must not submit a claim without selecting a terminal from a successful terminals response.

當航廈載入中時，顯示載入狀態，且不允許旅客在未選定航廈的情況下繼續。若載入失敗，顯示錯誤與重試動作。旅客不得在未從成功的航廈回應中選定航廈的情況下提交申請

Client-side claim quality score 客戶端的申請品質分數

To help passengers provide useful identifying information, calculate a **claim quality score** from 0 to 100 entirely in the browser. This is a client-side guidance feature: it must not call an API, must not be sent with the claim, and must not be presented as a probability that the item will be recovered.

為協助旅客提供有用的辨識資訊，完全在瀏覽器中計算一個 0 到 100 的**申請品質分數**。這是一項客戶端的引導功能：它不得呼叫 API、不得隨申請一併送出，也不得被呈現為物品能被尋回的機率

Recalculate the score immediately whenever a relevant form value changes:

每當相關表單值變動時，立即重新計算分數:

Rule	Points
A terminal is selected	10
A category is selected	10
Brand contains a non-whitespace value	15
Colour contains a non-whitespace value	10
Flight number contains a non-whitespace value	10
Character threshold 1: Trimmed description contains at least 40 characters	10
Character threshold 2: Trimmed description contains at least 80 characters	+10

Rule	Points
Description contains at least 12 distinct words, compared case-insensitively and ignoring punctuation	10
Date rule 1: lost_on is today or 1 day ago	15
Date rule 2: lost_on is 2–3 days ago	10
Date rule 3: lost_on is 4–7 days ago	5
Date rule 4: lost_on is 8 or more days ago, missing, invalid, or in the future	0

Only one date rule may contribute to the score. Character thresholds are cumulative, so a description of at least 80 characters receives 20 description-length points. Clamp the final result to the range 0–100.

只有一條日期規則可對分數有所貢獻。字元門檻為累計，因此至少 80 個字元的描述可獲得 20 分的描述長度分數。將最終結果夾限於 0 – 100 的範圍內

Display both the numerical score and its label:

同時顯示數值分數與其標籤:

Score	Label
0–39	Needs more detail
40–69	Basic
70–89	Detailed
90–100	Excellent

The review step must show a points breakdown and indicate which optional details could still increase the score. For example, a claim with a terminal and category, brand, colour, no flight number, an 85-character description containing

at least 12 distinct words, and today's date scores **90**.

檢視步驟必須顯示分數的細項拆解，並指出哪些選填細節仍可提高分數。例如，一筆具備航廈與類別、品牌、顏色、無航班編號、描述長 85 個字元且含至少 12 個相異單字、日期為今天的申請，得分為 ****90****

The score is advisory and must never prevent submission when the API-required fields are valid. When a draft is continued or discarded, the score must match the form values again.

此分數僅供參考，當 API 必填欄位皆有效時，絕不得阻止提交。當草稿被繼續或捨棄時，分數必須再次與表單值一致

After successful creation:

成功建立後:

- Show a **confirmation modal** with the newly created claim details from the API response. At least show the reference code (CL-XXXXXXXX), initial submitted status, category, terminal, and date lost. Make the reference code prominent and easy to copy.

顯示一個****確認彈窗****，內含 API 回應中新建立的申請詳情。至少顯示參考編號（`CL-XXXXXXXX`）、初始的 `submitted` 狀態、類別、航廈與遺失日期。讓參考編號醒目且易於複製

- The modal must offer exactly these two actions:

彈窗必須恰好提供以下兩個動作:

- **Create a new claim** — close the modal, reset the multi-step form to the first step, and leave the passenger ready to file another claim.

建立新申請****** — 關閉彈窗、將多步驟表單重設至第一步，並讓旅客準備好再提交另一筆申請

- **Show my claims** — navigate to the My claims page.

顯示我的申請****** — 導向「我的申請」頁面

- If the passenger dismisses the modal in any other way (Escape, a close button, or clicking the backdrop), treat that the same as **Show my claims**: navigate to the My claims page. Do not return them to the finished form

with no confirmation left on screen.

若旅客以任何其他方式關閉彈窗（Escape、關閉按鈕，或點擊背景遮罩），一律視同**顯示我的申請**：導向「我的申請」頁面。請勿把他們帶回已完成的表單，卻在畫面上沒有留下任何確認資訊

My claims page 我的申請頁

The My claims page route is /claims. It must retrieve only the authenticated passenger's claims.

「我的申請」頁面路由為 `/claims`。它必須只取回已驗證旅客本人的申請

- Display claims newest first as returned by the API.
依 API 回傳的順序，最新在前顯示申請
- Support the API's paginated response using its links or meta information.
Do not assume that all claims are returned on one page.
使用 API 的分頁回應，透過其 `links` 或 `meta` 資訊支援分頁。不得假設所有申請都會在同一頁回傳
- Show at least the reference code, category, terminal, date lost, and current status for each claim.
每筆申請至少顯示參考編號、類別、航廈、遺失日期與目前狀態
- Allow filtering by status.
允許依狀態篩選
- Each result must link to its claim detail page.
每筆結果都必須連結到其申請詳情頁

The claims page must behave as a **stateful claims workspace**. Store the selected status and page number in the URL query string (for example `?status=submitted&page=2`). Reloading the page, sharing its URL, and using browser back or forward must restore the same view. Changing the status must return to page 1.

申請頁必須表現為一個**具狀態的申請工作區**。將所選的狀態與頁碼儲存在 URL 查詢字串中（例如 `?status=submitted&page=2`）。重新載入頁面、分享其 URL，以及使用瀏覽器的上一頁或下一頁，都必須還原相同的檢視。變更狀態必須回到第 1 頁

Requests can finish in a different order when a passenger changes filters quickly. Cancel obsolete requests when possible, or otherwise ensure that an older response can never overwrite the latest selected view. Use the provided API's intentional delay on `?status=rejected` to verify this: select **rejected**, then quickly switch to another status (for example **resolved**) before the rejected response returns. After both requests have finished, the list and URL must still show the last selected status — not rejected.

當旅客快速切換篩選時，請求可能以不同的順序完成。盡可能取消過時的請求，否則須確保較舊的回應絕不會覆寫最新選定的檢視。請利用所提供 API 對 `?status=rejected` 的刻意延遲來驗證此點：選擇 **rejected**，然後在 rejected 回應返回前快速切換到另一個狀態（例如 **resolved**）。當兩個請求都完成後，清單與 URL 都必須仍顯示最後選定的狀態——而非 rejected

Keep the existing results visible with a non-blocking loading indication while moving between pages; do not briefly show an incorrect empty state.

在切換頁面時，保持既有結果可見並搭配非阻斷式的載入提示；不得短暫顯示不正確的空狀態

Claim detail and withdrawal page 申請詳情與撤回頁

The claim detail page route is `/claims/:id`. Retrieve a claim and present all useful fields returned by the passenger claim response, including:

申請詳情頁路由為 `/claims/:id`。取回一筆申請並呈現旅客申請回應中回傳的所有有用欄位，包含：

- Reference code
- Terminal
- Category
- Brand and colour when available
- Description
- Date lost
- Flight number when available
- Status

- Created and updated dates
- Resolution date when available
- Matched-item information when the API includes it

A passenger may withdraw their own claim only while its status is submitted or under-review.

旅客只有在申請狀態為 `submitted` 或 `under-review` 時，才可撤回自己的申請

- Show the withdrawal action only when the current status is eligible.
只在目前狀態符合資格時，才顯示撤回動作
- Ask for confirmation before sending the request.
在送出請求前要求確認
- On success, update the displayed claim to the returned withdrawn state.
成功時，將所顯示的申請更新為回傳的 `withdrawn` 狀態

Live claim journey 即時申請歷程

Present the claim's current state as a passenger-friendly journey rather than only displaying the raw status value. The journey must correctly represent the API statuses submitted, under-review, matched, resolved, rejected, and withdrawn. Rejected and withdrawn claims are alternative endings and must not be presented as successfully completed journeys.

將申請的目前狀態呈現為對旅客友善的歷程，而非只顯示原始的狀態值。此歷程必須正確表現 API 狀態 `submitted`、`under-review`、`matched`、`resolved`、`rejected` 與 `withdrawn`。被駁回與被撤回的申請屬於替代性的結局，不得被呈現為成功完成的歷程

Claim refreshing 申請重新整理

While an authenticated claim detail page is open, refresh it every **15 seconds**:
當一個已驗證的申請詳情頁開啟時，每 ****15 秒****重新整理一次:

- Do not start a new refresh while the previous refresh is still running.
在前一次重新整理仍在執行時，不得啟動新的一次重新整理
- Pause automatic refresh while the page is not visible and refresh immediately when it becomes visible again.

當頁面不可見時暫停自動重新整理，並在其再次變為可見時立即重新整理

- When the status changes, update the journey and claim information without a full page reload and announce the change visibly and through an accessible live region.

當狀態變更時，在不進行整頁重新載入的情況下更新歷程與申請資訊，並以視覺方式、以及透過可存取的即時區域宣告該變更

- Stop polling when the claim reaches resolved, rejected, or withdrawn.

當申請到達 `resolved`、`rejected` 或 `withdrawn` 時，停止輪詢

- Provide a **manual refresh** control. It must use the same loading and error handling as automatic refresh.

提供一個**手動重新整理**控制項。它必須使用與自動重新整理相同的載入與錯誤處理

- A temporary refresh failure must keep the last successful claim visible and show a non-destructive warning. Do not replace the claim with an empty or full-page error state. The manual refresh control may be used to retry.

一次暫時性的重新整理失敗必須保持最後一次成功的申請可見，並顯示一則非破壞性的警告。不得以空的或整頁的錯誤狀態取代該申請。可使用手動重新整理控制項來重試

Public claim tracking page 公開認領追蹤頁

The public claim tracking entry route is `/track`, and a tracking result uses `/track/:referenceCode`. Anyone must be able to track a claim without signing in by entering its reference code.

公開認領追蹤的進入路由為 `/track`，追蹤結果使用 `/track/:referenceCode`。任何人都必須能在不登入的情況下，藉由輸入參考編號來追蹤某筆申請

- Accept a reference code in the CL-XXXXXXXX format.

接受 `CL-XXXXXXXX` 格式的參考編號

- On success, navigate to the tracking result route and show only the public response data: reference code, status, category, terminal, date lost,

creation date, and resolution date when available.

成功時，導向追蹤結果路由，並只顯示公開回應資料：參考編號、狀態、類別、航廈、遺失日期、建立日期，以及結案日期（若有）

- Do not attempt to retrieve or display passenger personal data.
不得嘗試取回或顯示旅客個人資料
- A 404 response must produce a clear "claim not found" result and allow another search.

404 回應必須產生清楚的「找不到申請」結果，並允許再次搜尋

The reference code must appear in the tracking result route so that it can be bookmarked or shared. Reloading that URL must repeat the public lookup.

Normalize user input to uppercase and remove accidental surrounding whitespace before requesting it, but do not change the internal characters.

參考編號必須出現在追蹤結果的路由中，以便加入書籤或分享。重新載入該 URL 必須重複公開查詢。在請求前，將使用者輸入正規化為大寫並移除意外的首尾空白，但不得變更其內部字元

Apply the same live journey and 15-second refresh behaviour as the authenticated claim detail page. Public tracking must never call an authenticated passenger endpoint or reveal fields outside the public response. 套用與已驗證申請詳情頁相同的即時歷程與 15 秒重新整理行為。公開追蹤絕不得呼叫已驗證的旅客端點，也不得揭露公開回應以外的欄位

Passenger profile page 旅客個人資料頁

The profile page route is /profile. It must retrieve the authenticated passenger's profile details and allow updates.

個人資料頁路由為 `/profile`。它必須取回已驗證旅客的個人資料細節並允許更新

Display and allow editing of:

顯示並允許編輯:

- First name
- Last name

- Email address
- Phone number
- Address line 1
- Address line 2
- City
- Postcode
- Country

The passenger may also set a new password of at least 8 characters. Do not display an existing password and do not send an unchanged or empty password field.

旅客亦可設定一組至少 8 個字元的新密碼。不得顯示既有密碼，且不得送出未變更或空白的密碼欄位

The update endpoint supports partial updates. Show validation errors, including a non-unique email address, next to the relevant fields. After a successful update, show the returned current profile data and a visible success confirmation.

更新端點支援部分更新。將驗證錯誤（包含非唯一的 email 地址）顯示於相關欄位旁。成功更新後，顯示回傳的目前個人資料，以及一個可見的成功確認

Error handling 錯誤處理

Handle at least the following error cases throughout the application:

在整個應用程式中至少處理以下錯誤情況:

Status or condition	Required behaviour
401 Unauthenticated or invalid credentials	For a protected request, clear the invalid session and request sign-in. For the login form, show an authentication error.
404 Not Found	Explain that the claim or resource was not

Status or condition	Required behaviour
	found or is unavailable to this passenger.
422 Validation failed	Display the general message and field-specific errors.
422 business-rule failure	Display the API's descriptive message near the action that failed.
Network or unavailable API	Explain that the service could not be reached and allow the user to retry.

Prevent duplicate submissions while a request is in progress. Do not silently discard API errors.

在請求進行中時防止重複提交。不得默默地丟棄 API 錯誤

The application must prevent stale responses from updating a page after its route or authenticated passenger has changed. Automatic and manual refreshes, route changes, and sign-out must not leave timers or pending requests applying to the wrong page. Timers and pending requests must be cleaned up when their page is left.

應用程式必須防止過時的回應在其路由或已驗證旅客變更後仍更新頁面。自動與手動重新整理、路由變更與登出，都不得留下套用於錯誤頁面的計時器或待處理請求。當離開某頁面時，其計時器與待處理請求必須被清理乾淨

Design, responsiveness, and accessibility 設計、響應式與無障礙

The result must look and behave like a public airport passenger service, not an internal administration dashboard.

成果的外觀與行為必須像一個公開的機場旅客服務，而非內部的後台管理儀表板

- Use a clear visual hierarchy and readable status indicators. Status must not be communicated by colour alone.

使用清楚的視覺層級與易讀的狀態指示。狀態不得僅以顏色傳達

- Make forms usable with keyboard navigation and associate labels and error messages with their controls.
讓表單可透過鍵盤導覽操作，並將標籤與錯誤訊息與其控制項關聯
- Provide visible focus states and meaningful text for interactive controls.
為互動控制項提供可見的焦點狀態與有意義的文字
- Confirm destructive or consequential actions such as claim withdrawal.
對破壞性或有重大後果的動作（例如撤回申請）進行確認
- The application must remain usable without horizontal scrolling at common mobile and desktop viewport widths.
應用程式在常見的行動裝置與桌面視窗寬度下，都必須維持可用且不需水平捲動
- Use appropriate date, loading, and feedback presentation consistently across the portal.
在整個入口網站中一致地使用適當的日期、載入與回饋呈現方式

Assessment 評分

Module B will be assessed in the provided latest stable version of Google Chrome. Assessment will include:

模組 B 將在所提供的最新穩定版 Google Chrome 中評分。評分將包含:

- Correct integration with the provided Module A API
與所提供的模組 A API 正確整合
- Completeness and correctness of passenger workflows
旅客工作流程的完整性與正確性
- SPA routing, authentication persistence, and route protection
SPA 路由、驗證持久性與路由保護
- Multi-step form state, passenger-scoped draft recovery, and validation
多步驟表單狀態、依旅客範圍的草稿復原，以及驗證
- Correct client-side claim quality calculation, breakdown, and guidance
正確的客戶端申請品質計算、細項拆解與引導
- URL-driven pagination and filtering, including correct handling of

competing requests (verified with the delayed rejected filter)

由 URL 驅動的分頁與篩選，包含對競爭請求的正確處理（以延遲的`rejected` 篩選驗證）

- Live claim journeys, polling lifecycle, and status-change feedback
即時申請歷程、輪詢生命週期與狀態變更回饋
- Error handling and recovery from unavailable or stale API requests
錯誤處理，以及從不可用或過時 API 請求中的復原
- User experience, responsive behaviour, and accessibility
使用者體驗、響應式行為與無障礙
- Appropriate framework use and maintainable frontend code
適當地使用框架，以及可維護的前端程式碼

Any modification to the provided backend, its database, or its endpoint contract will not be considered during assessment. Only the documented consumer-facing operations listed in this task may be used.

對所提供後端、其資料庫或其端點契約的任何修改，在評分時都不予採計。只能使用本任務所列出、有文件記載的、面向消費者的操作

Mark distribution 分數分配

WSOS SECTION	Description	Points
1	Work organization and self-management	4
2	Communication and interpersonal skills	3
3	Design Implementation	8.5
4	Front-End Development	17.5
5	Back-End Development	0
Total		33